

The premise in 4 readings

The short path through the 60-source reading order for your infrastructural counterfactual mysteries: #1, #3, #12 and #14. Then the 3 key ideas the other 56 add up to.

1	The engine	Star (1999), <i>The Ethnography of Infrastructure</i>	p. 3
2	The thesis	David (1985), <i>Clio and the Economics of QWERTY</i>	p. 5
3	The city	Pierce (1995), <i>The Road to Lockport</i>	p. 7
4	The power fork	Jia & Crabtree (2015), <i>The International Niagara Commission of 1891</i>	p. 9
5	The other 56	The 3 key ideas from the rest of the reading order	p. 11

How to use it

- One reading a day, Monday to Thursday. Friday is the 3 ideas.
- Read each guide page first, then the paper, then work the questions on the back of the guide.
- The papers aren't printed here. All 4 sit behind paywalls (SAGE, JSTOR, ASHRAE and Cambridge), so each guide says how to get yours. Print it and slot it behind its guide.
- The guide summaries come from Claude's knowledge of each paper and its abstract. Claude hasn't read these 4 in full, so where a guide and the paper disagree, trust the paper.
- Print double-sided and each reading becomes one sheet.

How this week fits your engine

Your engine has 4 layers. Each reading and each idea in this packet feeds one of them.

1	The ordinary thing Traffic light, street address, elevator, refrigerator, telephone number, parking meter, house key, fire hydrant, crossing gate, thermostat.	The 10 items themselves
2	The fork, 1880-1960 A real decision that could have gone the other way.	Reading 4 · the Niagara Commission Idea 1 · The losing design worked
3	The accumulated world, 1962-82 The fork, built in and forgotten.	Reading 2 · lock-in Reading 3 · Lockport Idea 2 · The fork sinks into the walls
4	The mystery A breakdown that makes the system visible.	Reading 1 · the engine Idea 3 · The record and the machine drift apart

The week

DAY	READ	READING ORDER	LENGTH
Mon, Oct 5	Star (1999), <i>The Ethnography of Infrastructure</i>	#1	15 pages
Tue, Oct 6	David (1985), <i>Clio and the Economics of QWERTY</i>	#3	6 pages
Wed, Oct 7	Pierce (1995), <i>The Road to Lockport</i>	#12	Conference paper
Thu, Oct 8	Jia & Crabtree (2015), <i>The International Niagara Commission of 1891</i>	#14	16 pages
Fri, Oct 9	The 3 key ideas from the other 56, in this packet	The rest	5 pages

The engine

Star (1999), *The Ethnography of Infrastructure*

CITE	Star, Susan Leigh. 1999. "The Ethnography of Infrastructure." <i>American Behavioral Scientist</i> 43 (3): 377-391.
MAPS TO	The engine. Each mystery is a breakdown that makes a hidden system visible.
LENGTH	15 pages
GET IT	doi.org/10.1177/00027649921955326 (SAGE), through a library login. Most libraries can also get it for you by interlibrary loan.

The argument in brief

Star opens with "a call to study boring things": plugs, standards, forms. Her case is that the parts of modern life people rarely notice do a lot of the shaping.

Infrastructure, for Star, is relational. The same pipe is invisible to the person using it and the whole job of the person maintaining it.

She lists the properties that make something infrastructure. Two of them are your series in miniature: it's built on an installed base, and it becomes visible upon breakdown. Her method is ethnography: watch the work, read the documents, follow the people who keep a system running.

Mark these

- The opening call to study boring things. It's your collection's mission statement, written by a sociologist.
- The list of infrastructure properties. Pencil one of your stories next to each property.
- Every place she calls infrastructure relational. That's your license to rotate investigators, since each profession sees a different system.

Why it matters for the series

Every story in the collection is a breakdown, and every breakdown shows the reader a piece of the alternate world for a moment. Star gives you the reason the solve and the reveal can be one act: the system shows itself at the instant it fails.

Slot the paper behind this sheet. Questions are on the back.

Work these as you read

- 1 Which property on Star's list does each of your 10 ordinary items lean on hardest?

- 2 Who in your 1970s Lockport can see a system the residents can't? That person is a candidate investigator.

- 3 What's the boring document at the center of each story: a form, a standard, a log?

- 4 Where can a breakdown show the fork with no one explaining it?

Other notes

The thesis

David (1985), *Clio and the Economics of QWERTY*

CITE	David, Paul A. 1985. "Clio and the Economics of QWERTY." <i>American Economic Review</i> 75 (2): 332-337.
MAPS TO	The thesis. Civilization is made of forgotten decisions.
LENGTH	6 pages
GET IT	jstor.org/stable/1805621 , through a library login, or by interlibrary loan.

The argument in brief

Why you still type on QWERTY. A layout from the early typewriter business beat later rivals and stayed, because everything around it adapted to it.

David names 3 ingredients and calls the mix "QWERTY-nomics." Technical interrelatedness: the keyboard has to match what typists' fingers already know. Economies of scale: every new user makes the standard cheaper for everyone else. Quasi-irreversibility of investment: once millions of typists are trained, switching costs too much.

His point for historians is that sometimes the only way to understand why the world works the way it does is to learn how it got that way.

A small event worth knowing

On July 25, 1888, in Cincinnati, Frank McGurrian touch-typed on a QWERTY machine and beat Louis Traub, who was typing on a rival Caligraph. McGurrian took the \$500 prize, and the win helped make QWERTY and touch typing the norm. One contest, one typist, and a standard that has lasted more than a century.

Mark these

- The 3 ingredients. A fork has to carry all 3 to survive until 1962.
- Each place where a small, early event decides a long outcome.
- Where David treats the losing keyboard, Dvorak's, as better. Liebowitz & Margolis (#4) dispute that evidence. Hand their argument to your in-world skeptic.

Why it matters for the series

This is your thesis in the language of economics. The 3 ingredients explain why a city that bet on air and water mains in the 1890s still runs on them in 1973. The hardware, the skills and the money locked together, and the original argument faded from memory.

Slot the paper behind this sheet. Questions are on the back.

Work these as you read

- 1** For one story, name the fork's 3 ingredients: what's interrelated, what gets cheaper with scale, and what can't be undone.

- 2** What's your world's McGurkin moment, the small early event that tipped it?

- 3** Is their system better than ours, about equal, or worse? The answer sets how defensive your engineers sound.

- 4** What would it cost your Lockport to switch back in 1973, and who would pay?

Other notes

The city

Pierce (1995), *The Road to Lockport*

CITE	Pierce, Morris A. 1995. "The Road to Lockport: Historical Background of District Heating and Cooling." <i>ASHRAE Transactions</i> 101 (1).
MAPS TO	The setting, and why the collection lives in Lockport, New York. It also feeds The Dial, Cold Room and The Hydrant.
LENGTH	Conference paper
GET IT	Ask a library for <i>ASHRAE Transactions</i> vol. 101, part 1 (1995), or buy the paper from ASHRAE. The OSTI record at osti.gov/biblio/87450 has the citation.

The argument in brief

Pierce traces district heating, heat made in one plant and piped to many buildings, back to Roman hypocausts that warmed several buildings at once. The idea returns in the Renaissance and grows into hot-air, hot-water and steam heating, pushed along by the need to save fuel and cut smoke.

Then come the near misses. People proposed district systems in London (1623 and the 1820s), Pennsylvania (1869), and Warsaw and Zürich (1872). Steam ran at the US Naval Academy in 1853 and at the 1876 Philadelphia Centennial Exposition, and a hot-water cogeneration system ran at England's Banstead Downs Asylum the same year.

Then Birdsill Holly. In 1877 he installed the first successful commercial district heating system, in Lockport. By 1890 more than 50 were running, and many were still running when Pierce wrote.

Mark these

- Every early system that worked and then stalled. Each one is a fork you could flip.
- Wherever saving fuel and cutting smoke show up as reasons. That's your city's politics in the 1880s.
- Holly's 1877 system and the 50-plus by 1890. That's the head start your Lockport grows from. Arthur (#13) explains how a lead like that compounds.

Why it matters for the series

It's the documentary proof behind your setting. Lockport invented the district model. In our history the model won downtowns and lost the house. Your world lets it win both, and Holly's company becomes the nameplate on everything.

Slot the paper behind this sheet. Questions are on the back.

Work these as you read

- 1** What kept district heat out of houses in our history? Mark every reason Pierce gives.

- 2** Which early system would your 1970s Lockport put on its city seal?

- 3** Who owns the pipes in your city, a private company or the city itself?

- 4** If heat arrives by pipe, what does an ordinary 1973 apartment lack? A furnace, a chimney, a fuel bill? That's a missing object to leave unexplained.

Other notes

The power fork

Jia & Crabtree (2015), *The International Niagara Commission of 1891*

CITE	Jia, Jimmy Y., and Jason Crabtree. 2015. "The International Niagara Commission of 1891." In <i>Driven by Demand: How Energy Gets Its Power</i> , 93-108. Cambridge University Press.
MAPS TO	The power fork. The documented moment your world splits from ours.
LENGTH	16 pages
GET IT	doi.org/10.1017/CBO9781316221778.005 on Cambridge Core, through a library login, or borrow the book.

The event in brief

In 1890 the Cataract Construction Company, led by Edward Dean Adams, set up the International Niagara Commission to judge plans for putting the Falls to work and sending the power to Buffalo. Lord Kelvin chaired it. That September its secretary, W. C. Unwin, invited proposals.

The entries ran from pneumatic pressure to ropes, springs and pulleys, plus direct-current schemes. The commission rejected them all.

Alternating current won the contract in 1893 and reached Buffalo in 1896. Read the chapter for how the commission reached its verdict and who pushed for what.

Mark these

- Who sat on the commission, and what each one favored. Those are your fork's characters.
- Every proposal the commission took seriously before rejecting it. The pneumatic one is your world.
- The reasons given against air and water. Your fork has to flip only the weakest one.

Why it matters for the series

This is where your world splits from ours. Air and water mains work across a city and lose too much over long distances, so in your history Niagara's power stays near Niagara, industry climbs the escarpment, and Lockport becomes the center of a metro.

If the commission hooks you, read Lubar (1989, #15) next. He argues the choice rested on engineers' cultural assumptions as much as on cost.

Slot the paper behind this sheet. Questions are on the back.

Work these as you read

- 1** What's the smallest change that makes the commission back air or water: a different chair, a patent fight, a failed demonstration?

- 2** Who loses in your world? Name a company, a city and a person from our history.

- 3** How far can air or water power travel before losses eat it? That distance draws your metro's edge.

- 4** What does a Lockport schoolkid learn about 1891, in 1973? That's the official story your city tells itself.

Other notes

The 3 key ideas from the other 56

The 4 readings give you the engine, the thesis and the city. The other 56 sources in the reading order add up to 3 ideas, one each for layers 2, 3 and 4 of your engine. Every one of the 56 sits under exactly one idea. Here they are at a glance, then one page each. Numbers like #30 point to the reading order.

1

LAYER 2 · THE FORK · 17 SOURCES

The losing design worked

Every fork in the collection was a live contest between working machines. Money, institutions and accidents settled it.

PLAYBOOK RULE Working loser, smallest rewrite.

2

LAYER 3 · THE ACCUMULATED WORLD · 10 SOURCES

The fork sinks into the walls

Once a system wins, institutions grow around it and it drops out of sight, except for the people who keep it running.

PLAYBOOK RULE Let the paperwork show the fork.

3

LAYER 4 · THE MYSTERY · 29 SOURCES

The record and the machine drift apart

Most mechanisms in the corpus are a record that disagrees with a physical system. The solve reads the machine with a tool that existed by 1982.

PLAYBOOK RULE Log versus machine.

Friday notes

The losing design worked

Every fork in the collection was a live contest between working machines. Money, institutions and accidents settled it.

The corpus is full of machines that worked and lost. The quiet gas-absorption fridge lost to the compressor fridge because GE, GM, Westinghouse and the electric utilities had the capital and the motive (#30). Bell kept human operators for decades because they were cheap, flexible and part of the service (#50), and some towns ran 2 unconnected phone networks side by side (#51).

Paternosters moved 60 people per 5 minutes against 40 for conventional lifts in a 2008 Sheffield survey, and safety concerns stopped new ones from being built (#43). Kent Automatic Garages parked about 1,000 cars in one Manhattan tower (#46), and San Francisco built separate high-pressure fire mains after 1906 (#18).

Air and water did real work. Paris ran its clocks on pulses of air through street mains (#22). A hydraulic air compressor at Ragged Chutes, Ontario, made compressed air from falling water for about 70 years (#16), and fluidic logic built from jets of air boomed through the 1960s (#23).

Accident and taste did the deciding. London's first traffic signal blew up in 1869, and London waited until the 1920s to try again (#34). Engineers wanted whole-building “manufactured weather,” and consumers bought window units (#26). William Love broke ground on a power-canal city in 1894 and lost his investors by 1897, once power could travel (#17).

Designs stay open until the argument around them closes (#5). The Niagara argument closed partly on engineers' cultural assumptions (#15), and small early leads compound (#13). One guardrail: markets do switch when an alternative is clearly better (#4), so a fork survives to 1962 only if the winner was at least good enough. Change as little as possible, then follow the knock-on effects honestly (#6).

Where it shows up

Layer 2 of every story. Strongest in Cold Room, The Exchange, The Lift and The Meter, where the losing machine has the best paper trail.

Playbook rule

Working loser, smallest rewrite. For each story, write down the historical decision, its year, who made it, and the smallest change that flips it. If our winner was clearly better, pick another fork.

From the reading order: #4 Liebowitz & Margolis 1990 · #5 Pinch & Bijker 1984 · #6 Lebow 2000 · #13 Arthur 1989 · #15 Lubar 1989 · #16 Millar 2014 · #17 Newman 2016 · #18 Scawthorn et al. 2006 · #22 Popular Science Monthly 1882 · #23 Kirshner 1969 · #26 Cooper 1998 · #30 Cowan 1985 · #34 McShane 1999 · #43 Bottomley 2014 · #46 New York Times 2014 · #50 Lipartito 1994 · #51 Mueller 1993

The fork sinks into the walls

Once a system wins, institutions grow around it and it drops out of sight, except for the people who keep it running.

Hughes calls it momentum: big systems gather people, capital and rules until they're hard to turn (#7). Unruh shows lock-in covers institutions along with hardware: utilities, codes, unions, lenders (#8). Edwards shows mature infrastructure becoming the natural background of modern life (#2).

Defaults have birthdays. The national 68°F thermostat target dates to a November 1973 speech (#27). House numbers and city directories were built to make cities readable to government and business (#39). Minimum parking requirements remade American land and street life (#47).

Our world eliminated grade crossings case by case over decades (#54). Yours finished the job, so rail above or below every road is simply how a city looks. Blacklock's essay on riding a paternoster shows how fast a strange machine starts to feel ordinary (#45).

Only some people can see the system. A pipe is invisible to a tenant and central to its crew (#9). Russell & Vinsel argue technology history should be told through upkeep (#11). Your investigators come from those crews, and your scratched, stickered look comes from the same place: maintenance is where a system shows.

Where it shows up

Layer 3 of every story, and the rule that characters never explain the missing object. Strongest in The Number, The Meter, The Dial and The Crossing.

Playbook rule

Let the paperwork show the fork. Give each story one institution that exists only because of the fork (a union local, a code, a utility, a standard) and one default with a birthday. Then cast someone who works on the system as the investigator.

From the reading order: #2 Edwards 2003 · #7 Hughes 1969 · #8 Unruh 2000 · #9 Star & Ruhleder 1996 · #11 Russell & Vinsel 2018 · #27 Nixon 1973 · #39 Rose-Redwood 2008 · #45 Blacklock 2020 · #47 Shoup 1999 · #54 Ogden & Cooper 2019 (FHWA)

The record and the machine drift apart

Most mechanisms in the corpus are a record (a log, a map, a plan) that disagrees with a physical system. The solve reads the machine with a tool that existed by 1982.

THE RECORD SAYS	THE MACHINE DOES	
The survey closes	Two datums put the same point in 2 places	#36
This address exists	Broken coordinates collapse to (0, 0) and become a place	#40
The house is here	Address matching lands it tens of meters off	#41
These records are one person	A false match invented them	#42
Everyone who entered left	The count was taken by flow at one landing	#44
The car hasn't moved in 27 years	The garage shuffles it to speed retrieval	#48, #49
The line is disconnected	A bridged tap keeps it on the wire	#52
The shift was quiet	It was staffed for calls the logs don't show	#53
The valve is closed	It sits open and joins 2 pressure zones	#19
Someone came in	A pressure spike flipped the latch	#25
Only the master opens every door	One tenant's key reveals the master	#24

Physics supplies the rest. A weave puts crashes where the plan shows none (#35), and ethylene ripens a whole shared cold store (#31). Night inversions carry a train's sound, and rock carries its rumble (#55, #56). A heat pipe holds a set point with no power (#29), and any steady temperature implies a governor somewhere (#28).

Period tools settle it: water isotopes (#20), bomb tritium (#21), the sound spectrograph (#58), an ethylene detector (#33), Baker's crash manual (#38) and the Hardy Cross method (#19). Two traps keep it honest. Niagara dolostone doesn't swell in concrete (#37), and temperature alone speeds decay only 2 to 3 times per 10°C (#32).

And the answer is often a system. The famous story about Robert Moses's low bridges turns out to be mostly legend (#10), and the Windsor Hum came from blast furnaces across a border (#57). That's your precedent for endings with no culprit.

Where it shows up

Layer 4 of every story. The record gap carries The Number, The Meter, The Lift, The Exchange and The Lock. Physical traces carry The Hydrant, The Crossing, Cold Room and The Dial. The Intersection gets both.

Playbook rule

Log versus machine. Write each mystery as a disagreement between a record and a physical system. The investigator wins by reading the machine with a tool from 1982 or earlier.

From the reading order: #10 Winner 1980 and Joerges 1999 · #19 Cross 1936 · #20 Craig 1961 · #21 Begemann & Libby 1957 · #24 Blaze 2003 · #25 Walters & Leishear 2018 · #28 Maxwell 1868 · #29 Grover et al. 1964 and Marcus 1972 · #31 Burg & Burg 1962 · #32 Labuza 1984 · #33 Burg & Stolwijk 1959 · #35 Pulugurtha & Bhatt 2010 · #36 Dewhurst 1990 · #37 Rogers et al. 2000 · #38 Baker 1975 · #40 Juhász & Mooney 2022 · #41 Zandbergen 2008 · #42 Fellegi & Sunter 1969 · #44 Fruin 1974 · #48 Roodbergen & Vis 2009 · #49 Li & Miao 2020 · #52 Reeve 2009 · #53 Erlang 1917 · #55 Embleton 1996 · #56 Colaço et al. 2025 · #57 Windsor Hum study 2014 · #58 Koenig et al. 1946

Put together

The 3 ideas give each story a spine you can check in 3 lines.

1	The fork	A working loser, and the smallest rewrite that flips it.
2	The world	One institution and one default that exist only because of the fork.
3	The mystery	A record that disagrees with the machine, and a tool from 1982 or earlier that settles it.

We're lucky the history is this rich. Almost every fork you want has a working loser waiting in the archive, with dates, prices and names attached.

One caution. The corpus can carry a lot of weight, so let it carry the fairness of the ending and keep it out of the dialogue. If a scene needs the reader to understand the physics, the physics is doing a character's job.

Next week: #2, #4, #5 and #6 from the reading order, the next 4 on the spine. Then pick the story you'll draft first and read its block.

This week's citations, for library requests

David, Paul A. 1985. "Clio and the Economics of QWERTY." *American Economic Review* 75 (2): 332-337. JSTOR 1805621.

Jia, Jimmy Y., and Jason Crabtree. 2015. "The International Niagara Commission of 1891." In *Driven by Demand: How Energy Gets Its Power*, 93-108. Cambridge University Press. doi:10.1017/CBO9781316221778.005.

Pierce, Morris A. 1995. "The Road to Lockport: Historical Background of District Heating and Cooling." *ASHRAE Transactions* 101 (1). OSTI 87450.

Star, Susan Leigh. 1999. "The Ethnography of Infrastructure." *American Behavioral Scientist* 43 (3): 377-391. doi:10.1177/00027649921955326.

All 158 sources, with notes: [story-research/01-science-corpus.md](#) and [corpus.bib](#) in the Orchestration repo.